

Structure Databases

Shatakshi Kulkarni

Introduction

- ❑ Structure databases are specialized repositories that store and provide access to the 3D structures of biological macromolecules, such as proteins, nucleic acids, and complex assemblies.
- ❑ These databases are crucial for understanding the function, dynamics, and interactions of biomolecules.
- ❑ Applications:
 - Finding good drug candidate by getting to know the biological activity through structure
 - Provide structures to further help in the identification of drug targets, and the design of inhibitors and other therapeutic agents
 - Provide reference for predicting structures of a sequence (Molecular modeling)
 - Structural databases often link to other biological databases, integrating structural data with functional, genomic, and proteomic information.

PDB/Protein Data Bank

- ❑ <https://www.rcsb.org/>
- ❑ The PDB has been the central repository for biomolecular structures since 1971, providing free access to 3D structures for scientific exploration.
- ❑ The website hosts advanced technologies that support efficient query and reporting of PDB data, including auto-complete search suggestions and faceted browsing options.
- ❑ Users can browse for protein which can be a drug target, drug candidate and ligands. E.g., hemoglobin
- ❑ Build complex queries using the Advanced Search feature, search for ligands by various criteria, and access data integrated with external resources for comprehensive exploration.
- ❑ Provides insights into how molecular machines function can aid in the development of targeted therapies for diseases by identifying specific molecular targets within cells
- ❑ Molecular machines play crucial roles in various cellular functions, such as cutting food, building molecules, supporting cells, and defending against infections
- ❑ Useful in Understanding the intricate molecular machinery within cells can lead to advancements in various fields, such as medicine, biotechnology, and bioengineering
- ❑ PDB structure database can be used as a reference database for Homology Modeling i.e., predicting a structure of a sequence.

PDB/Protein Data Bank

```
# Number of sequences: 30
# Length of profile : 335
# N_PROF ITERATIONS : 1
# GAP PENALTIES_1D : -900.0 -50.0
# MATRIX_OFFSET : 0.0
# RR_FILE : ${MODINSTALL8v1}/modlib/as1.sim.mat
1 TvLDH S 0 335 1 335 0 0 0 0. 0.0
2 1a5z X 1 312 75 242 63 229 164 28. 0.83E-08
3 1b8pA X 1 327 7 331 6 325 316 42. 0.0
4 1bdmA X 1 318 1 325 1 310 309 45. 0.0
5 1t2dA X 1 315 5 256 4 250 238 25. 0.66E-04
6 1civA X 1 374 6 334 33 358 325 35. 0.0
7 2cmd X 1 312 7 320 3 303 289 27. 0.16E-05
8 1o6zA X 1 303 7 320 3 287 278 26. 0.27E-05
9 1ur5A X 1 299 13 191 9 171 158 31. 0.25E-02
10 1guzA X 1 305 13 301 8 280 265 25. 0.28E-08
11 1gv0A X 1 301 13 323 8 289 274 26. 0.28E-04
12 1hyeA X 1 307 7 191 3 183 173 29. 0.14E-07
13 1i0zA X 1 332 85 300 94 304 207 25. 0.66E-05
14 1i10A X 1 331 85 295 93 298 196 26. 0.86E-05
15 1ldnA X 1 316 78 298 73 301 214 26. 0.19E-03
16 6ldh X 1 329 47 301 56 302 244 23. 0.17E-02
```

RCSB PDB Deposit Search Visualize Analyze Download Learn About Documentation Careers COVID-19 MyPDB Contact us

222,415 Structures from the PDB
1,068,577 Computed Structure Models (CSM)

Enter search term(s), Entry ID(s), or sequence
Include CSM ☐

Advanced Search | Browse Annotations | Help

PDB-101 PDB PDBDataBank NAKB RCSB PDB PDB-Dev

Structure Summary Structure Annotations Experiment Sequence Genome Ligands Versions

Biological Assembly 1

1X8V

Estriol-bound and ligand-free structures of sterol 14alpha-demethylase (CYP51)

PDB DOI: <https://doi.org/10.2210/pdb1X8V/pdb>

Classification: OXIDOREDUCTASE
Organism(s): Mycobacterium tuberculosis
Expression System: Escherichia coli
Mutation(s): Yes

Deposited: 2004-08-18 Released: 2004-11-23
Deposition Author(s): Podust, L.M., Yermakitskaya, L.V., Lepesheva, G.I., Podust, V.N., Dalmasso, E.A., Waterman, M.R.

Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 1.55 Å
R-Value Free: 0.227
R-Value Work: 0.206

wwPDB Validation

Metrics Percentile Ranks Value
R-Value 0.227 0.219
Checkscore 1.0 1.0
Ramachandran outliers 0.0 0.0

Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (HEM)

Small Molecules				
Ligands 2 Unique				
ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
HEM Query on HEM Download Ideal Coordinates CCD File Download Instance Coordinates	B [auth A]	PROTOPORPHYRIN IX CONTAINING FE C ₃₄ H ₃₂ Fe N ₄ O ₄ KABFMIBPWCXCRK-RGGAHWMASA-L		Interactions Interactions & Density
ESL Query on ESL Download Ideal Coordinates CCD File Download Instance Coordinates	C [auth A], D [auth A]	ESTRIOL C ₁₈ H ₂₄ O ₃ PROQIPRRNZUXQM-ZXXIGWHRSA-N		Interactions Interactions & Density

NDB/Nucleic acid Database

- ❑ <https://www.nakb.org/>
- ❑ The NDB was established in 1991 to compile and distribute structural information about nucleic acids, evolving to include RNA structures and specialized tools for the nucleic acid community.
- ❑ Easier for users to access and analyze nucleic acid structures and its biological roles effectively
- ❑ Various search capabilities, classifications of structures and motifs, and tools for nucleic acid analysis
- ❑ Sections in NDB and their roles:
 - **About section:** Info about the team, Citation of NDB references
 - **Standard:** Ideal geometry of DNA/RNA bases, Sugar phosphate backbone (Angle)
 - **Tools:** Based on functionality annotation trees enhances search of RNA or DNA



NDB/Nucleic acid Database

□ Interpretation of results:

- Sequence Information: The nucleotide sequence of the DNA or RNA, along with any associated protein sequences.
- Experimental Data: Information about how the structure was determined, such as the resolution of an X-ray crystallography experiment.
- Annotations and Functional Info: Information on recognized structural motifs and functional domains within the molecule.
- Visualize the secondary structure of RNA.

NAKB Protein/DNA **PDB 8P5E** View Download Tools Help

Released 2024-05-29 **Deposited** 2023-05-24

Title S. cerevisiae nexus-sCMGE after DNA replication initiation

Authors Henrikus SS, Willhoft O

Method ELECTRON MICROSCOPY 3.9 Å **EMDB** SINGLE PARTICLE Software: PHENIX

Primary Citation Unwinding of a eukaryotic origin of replication visualized by cryo-EM. Henrikus SS, Gross MH, Willhoft O, Puhringer T, Lewis JS, McClure AW, Greiwe JF, Palm G, Nans A, Diffley JFX, Costa A *Nat Struct Mol Biol* (2024) **PubMed** **DOI**

Entry Content Deposited MW: 1140.44 kDa

Analysis **DSSR** **DNATCO**

Protein Annotations [helicase](#) [polymerase](#) [cell cycle](#) [DNA replication/repair](#) [DNA metabolism protein](#) [DNA-directed DNA polymerase](#) [replication origin binding protein](#) [general transcription factor](#)

Components **Assemblies (1)** **Images**

Type	Description (hover for sequence)	Chains(AUTH)	Source	MW(kDa)	Links
DNA	DNA (19-MER)	A	synthetic construct	5.9	
	DNA (7-MER)	B	synthetic construct	2.1	



SCOP/Structural Classification of Proteins

- ❑ SCOP aims to organize proteins based on their structural and evolutionary relationships, grouping them into families, superfamilies, folds, and classes. It is a hierarchical classification of proteins.
- ❑ The classification in SCOP is based on careful analysis of protein structures, function, and evolution.
- ❑ Provides understanding of protein evolution, structure prediction, genome analyses, and more.
- ❑ Browse proteins or search with the help of option of browse by structural class and browse by protein type.
- ❑ Narrow the search till you get to the domain. This will help you to analyze the evolutionary hierarchical protein classification.
- ❑ The domain result provides sequence of that particular narrowed down domain.
- ❑ This sequence can be further used for sequence alignment, homology modeling, or identifying similar domains in other proteins.



SCOP/Structural Classification of Proteins

❑ Interpretation of results/What to expect from SCOP:

- How proteins are grouped into families, superfamilies, and folds based on their structural similarities and evolutionary origins
- Structural features, evolutionary relationships, and functional annotations
- Compare different proteins within the same family, superfamily, or fold to understand their similarities and differences in structure and function
- Classify and predict domain structures and functions based on sequence and structural similarities.

The screenshot shows the SCOP database interface. At the top, there's a navigation bar with the SCOP logo, 'About', 'Contact', and 'Download' links, and a search bar containing 'hemoglobin'. Below this, a 'FAMILY' box identifies the entry as 'Extracellular hemoglobin linker subunit, receptor domain'. The main 'DOMAIN' box displays '2GTL M:102-225' and 'SCOP ID: 8029676', along with 'Hemoglobin linker chain L1' and 'Species: Lumbricus terrestris'. A 'Show ancestry' checkbox is present. At the bottom, a 'Representative sequence' box shows a sequence alignment for 'UniProtQ9GV76' and 'This domain'.

← → ↻ ebi.ac.uk/pdbe/scop/term/8029676 ☆ 👤

About Contact Download hemoglobin

FAMILY Extracellular hemoglobin linker subunit, receptor domain

DOMAIN
2GTL M:102-225 SCOP ID: 8029676
Hemoglobin linker chain L1
Species *Lumbricus terrestris*
Show ancestry ☐

Representative sequence Other domains for this sequence External links

UniProtQ9GV76
This domain

IMVYLGAGLY GLAAGASDPY QSRPFQIVR NQNLSDYLA KRLADIRREY NKLTHDVKR TIRGLKARIS KLEENRDER ESECRGVPE CIRDLFCDS EDCRDSDE SPETCEINLT HVGSSTYLA TWYSCDIAIP DVAIVTITAA KRSFFPNQW KLAATLSTEL
DSRHTYVTT QLAQTFNQR KELLAPLAG QSRHTDVICD PHGECGSD CRIVPSSLP VMHPNQRK

CATH

- ❑ CATH is a hierarchical classification of protein domain structures. It categorizes protein domains based on their structural and evolutionary relationships.
- ❑ The CATH hierarchy consists of five major levels:
 - Class (C-level): broad classification based on secondary structure i.e., alpha helix, beta sheets
 - Architecture (A-level): description of the gross arrangement of secondary structures, independent of connectivity
 - Topology (T-level): structures with the same overall fold, indicating similar arrangements of secondary structures but may have diverse functions
 - Homologous superfamily (H-level): highly similar structures and functional similarity indicating evolutionary origin
 - Sequence family (S-level): significant sequence similarity and thus a high probability of having similar structure/function

CATH

CATH

HomeSearchBrowseDownloadAboutSupport

Search CATH by keywords or ID

CATH Domain 3bcqA00

1 keywords

Home / Superfamily 1.10.490.10 / Domain 3bcqA00

DOMAIN LINKS

Summary

Structure

Sequence

Neighbourhood

CATH Classification

Level	CATH Code	Description
1	1	Mainly Alpha
2	1.10	Orthogonal Bundle
3	1.10.490	Globin-like
4	1.10.490.10	Globins

Domain Context

PDB Structure

PDB 3BCQ

External Links

- PDBSum
- Proteopedia

MethodX-RAY DIFFRACTION

Organism

Primary Citation

Crystal structure of oxy-hemoglobin from Brycon cephalus

Poy, C.D., Leopoldino, A.M., Rahal, P., de Azevedo, W.F., Rodriguez, G.O.B., Murakami, M.T.

To be Published

Browse CATH-Gene3D Hierarchy

BROWSE LINKS

Browse Hierarchy

Highly Diverse Superfamilies

Superfamily Comparison

Select a CATH node...

Helix Hairpins

1.10.287

CATH ID1.10.287

Superfamilies153

Domains6538

Example Domain3buB00 (PDB)

Top of CATH hierarchy

1 Mainly Alpha

1.10 Orthogonal Bundle

1.10.287 Helix Hairpins

Class

Architecture

Topology

Homologous Superfamily

CSD/Cambridge Structural Database

- ❑ <https://www.ccdc.cam.ac.uk/>
- ❑ Repository of experimentally determined small molecule structures
- ❑ The data, typically obtained by X-ray crystallography and less frequently by electron diffraction or neutron diffraction, and submitted by crystallographers and chemists from around the world, are freely accessible (as deposited by authors) on the Internet via the CSD's parent organization's website (CCDC, Repository).
- ❑ Contains detailed information about the 3D structures of molecules, including atomic coordinates and crystallographic data.
- ❑ Can measure distance, bond angle, torsions for understanding molecular conformations
- ❑ Can investigate the interactions between molecules in the crystal lattice, such as hydrogen bonding, van der Waals interactions.
- ❑ CSD files can be exported in a wide range of formats, including CIF, MOL, Mol2, PDB, SHELX and XMol

CSD/Cambridge Structural Database

ccdc.cam.ac.uk/structures/Search?Compound=aspirin&DatabaseToSearch=Published

CCDC FIZ Karlsruhe
Leibniz Institute for Information Infrastructure

CSD Entry: ACMEBZ [Sign In](#)

Simple Search Structure Search Unit Cell Search Formula Search

Your query was: Compound name: aspirin and the search returned more than 30 records. [Back to Search List](#) [Modify Search](#) [New Search](#)

Results

☒ Database Identifier	☐ Deposition Number
☒ ACMEBZ	1100773

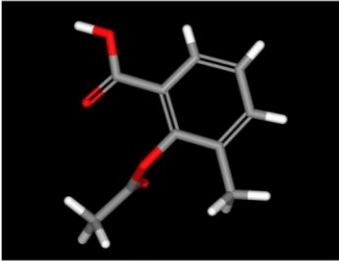
[Next](#)

[Download](#)

ACMEBZ : 2-Acetoxy-3-methylbenzoic acid
Space Group: $P 2_1/c$ (14), Cell: a 4.910(1)Å b 11.702(2)Å c 17.233(6)Å, α 90° β 98.36(2)° γ 90°

3D viewer

Capped Sticks ▼ No Labels ▼



Chemical diagram

